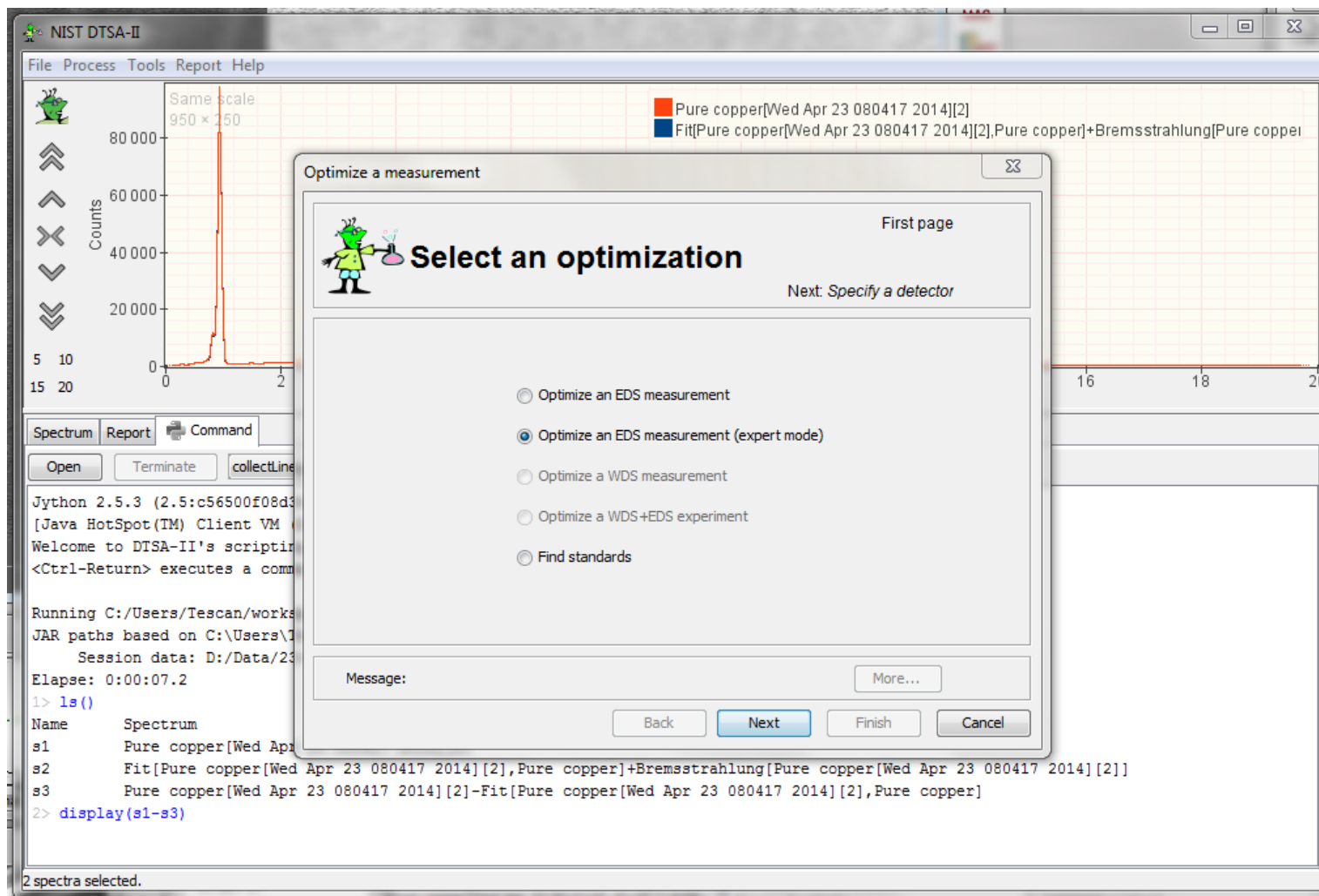
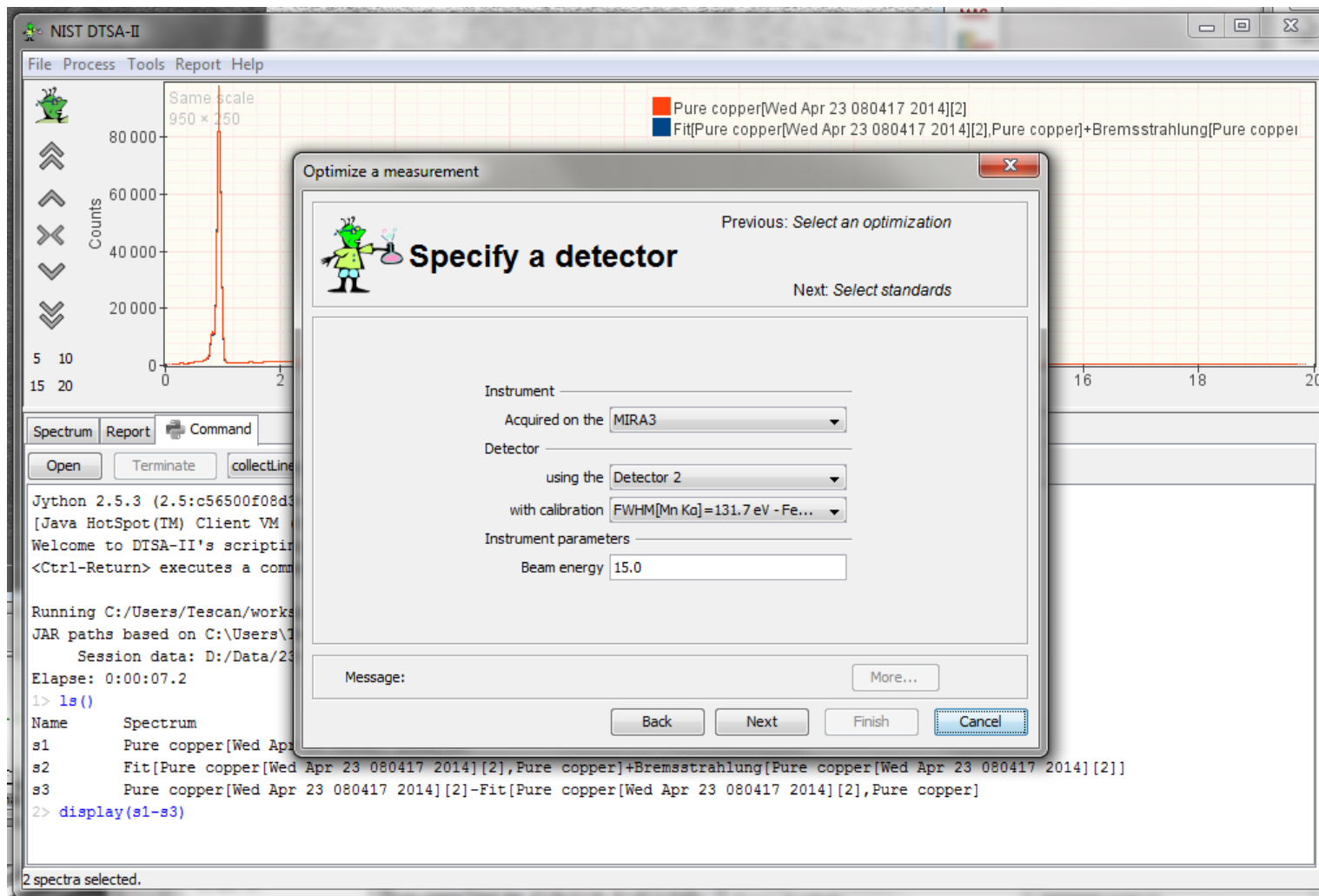
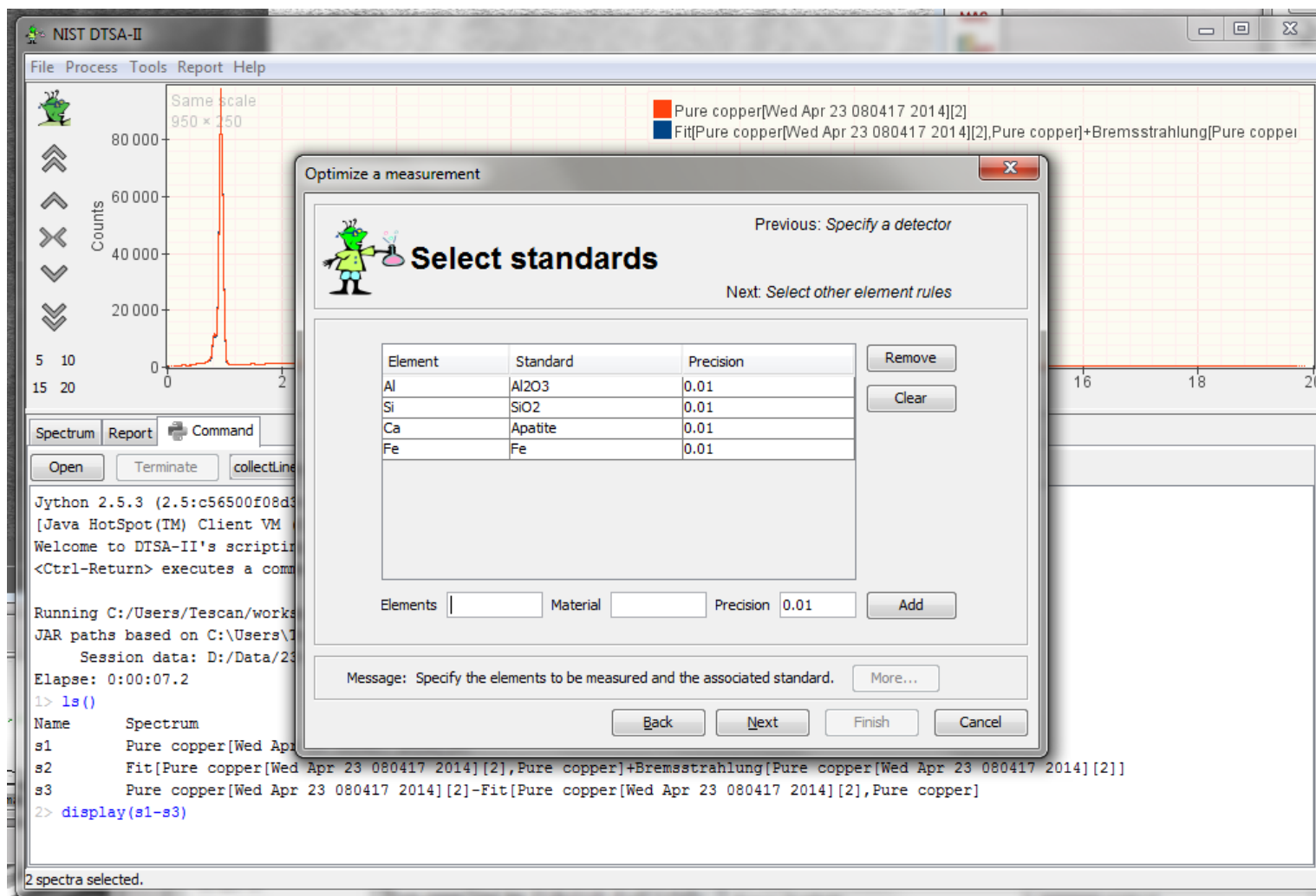


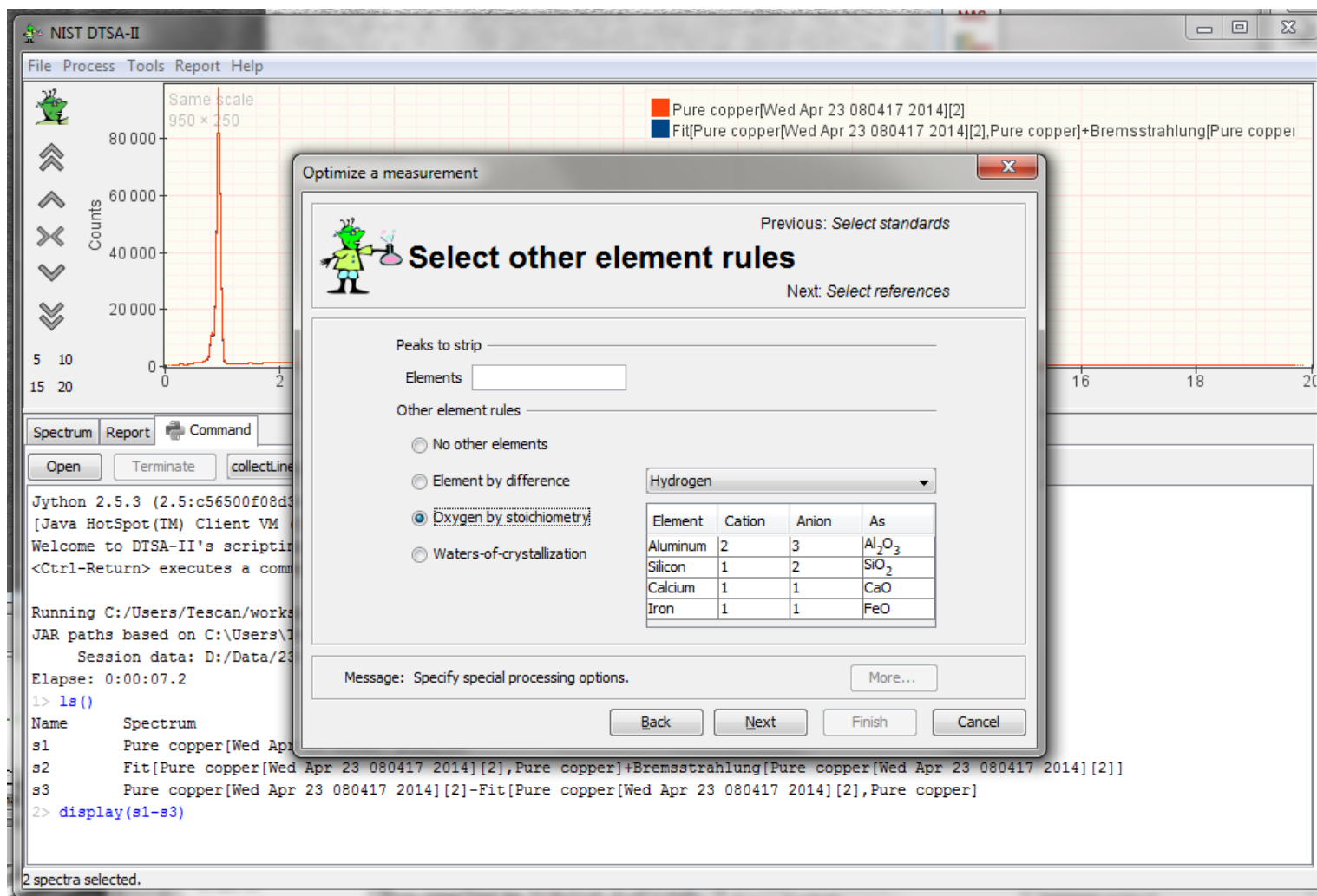
Optimization

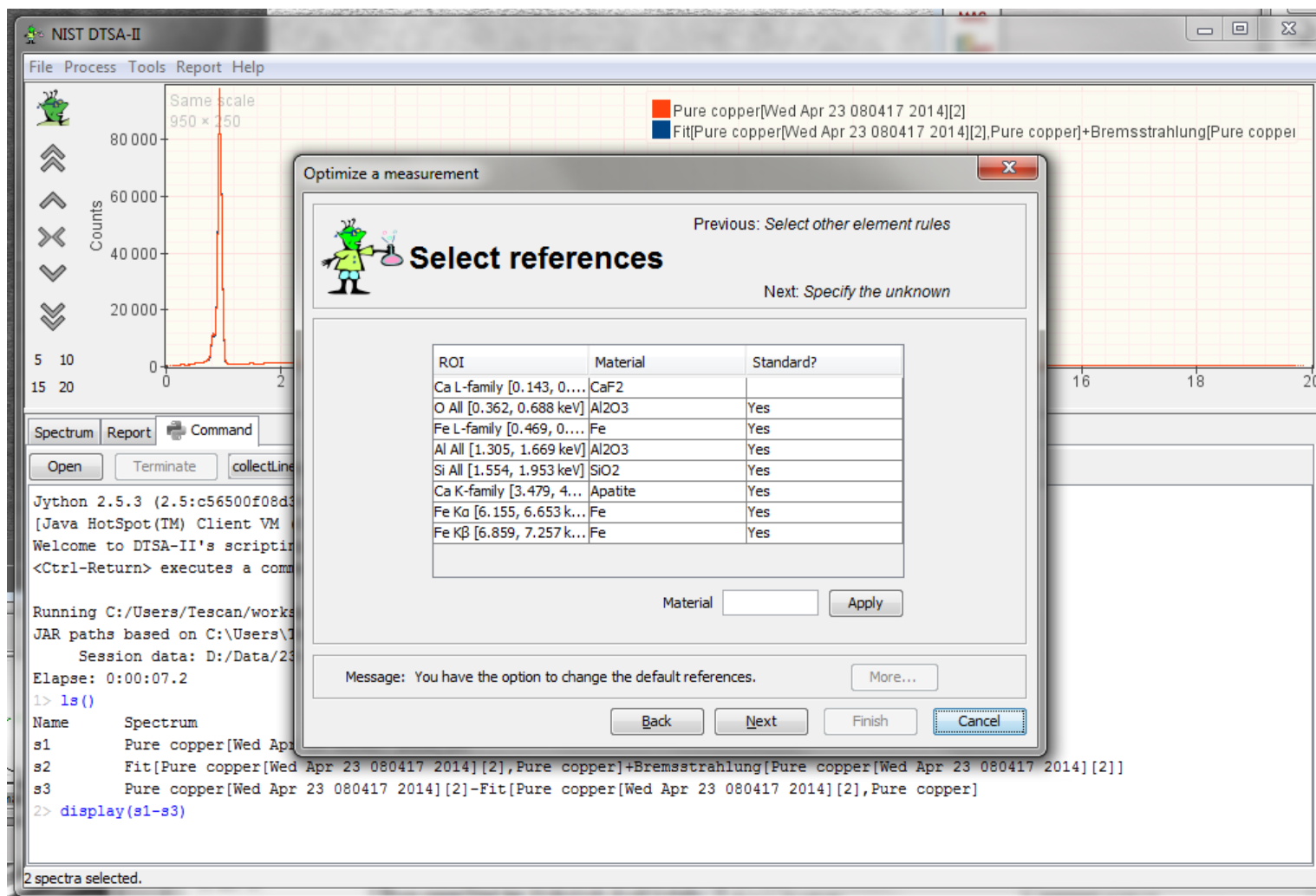
Designing a measurement protocol to optimally address the measurement need

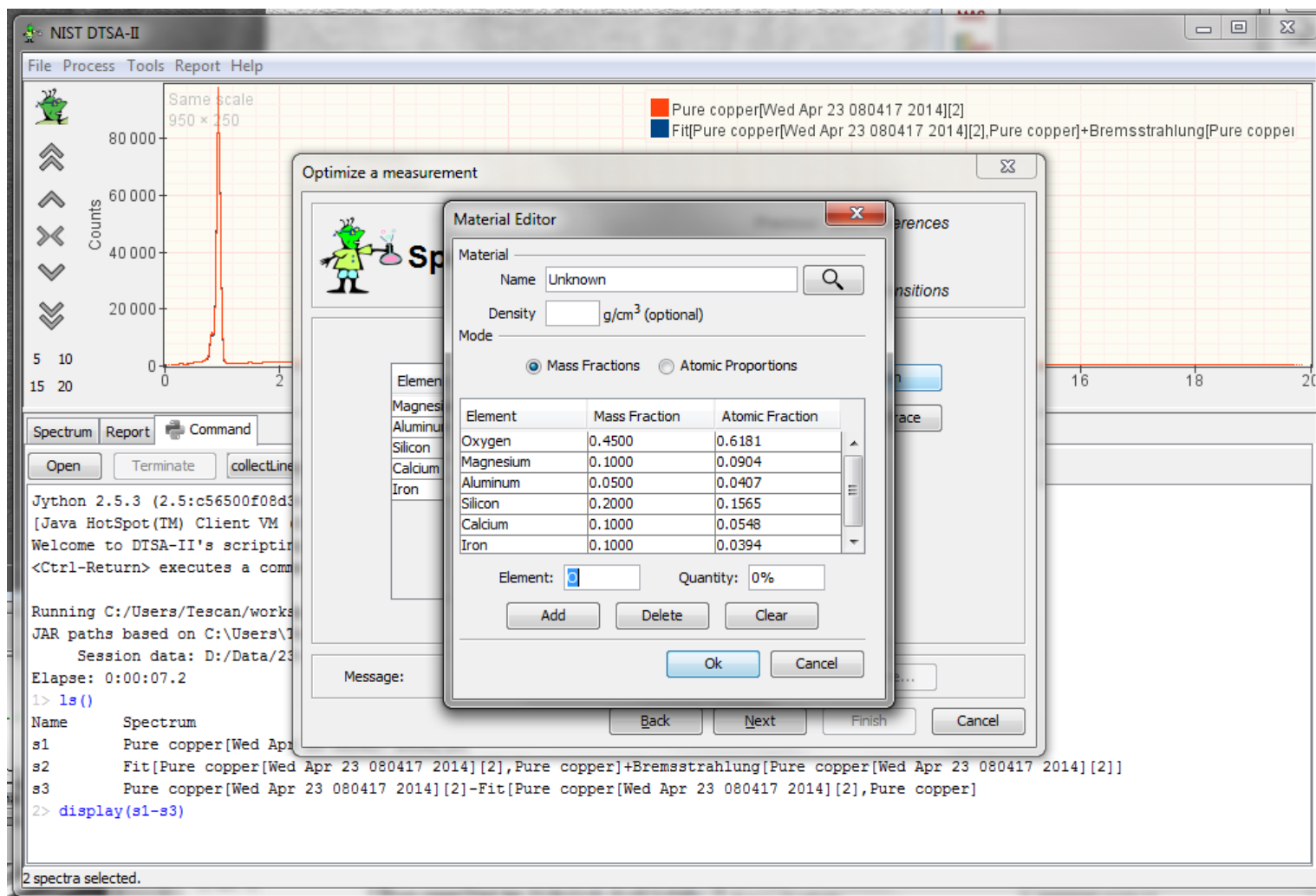


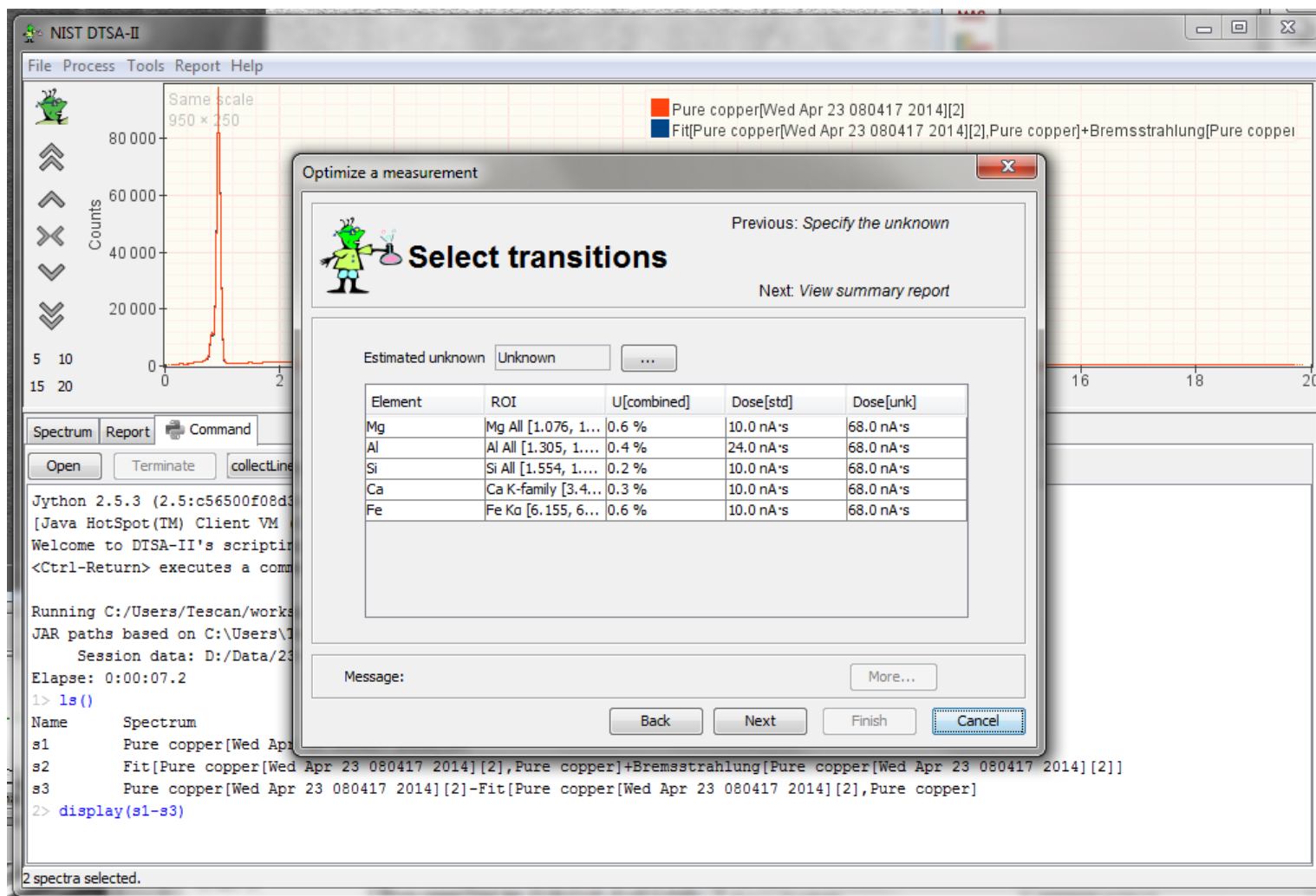


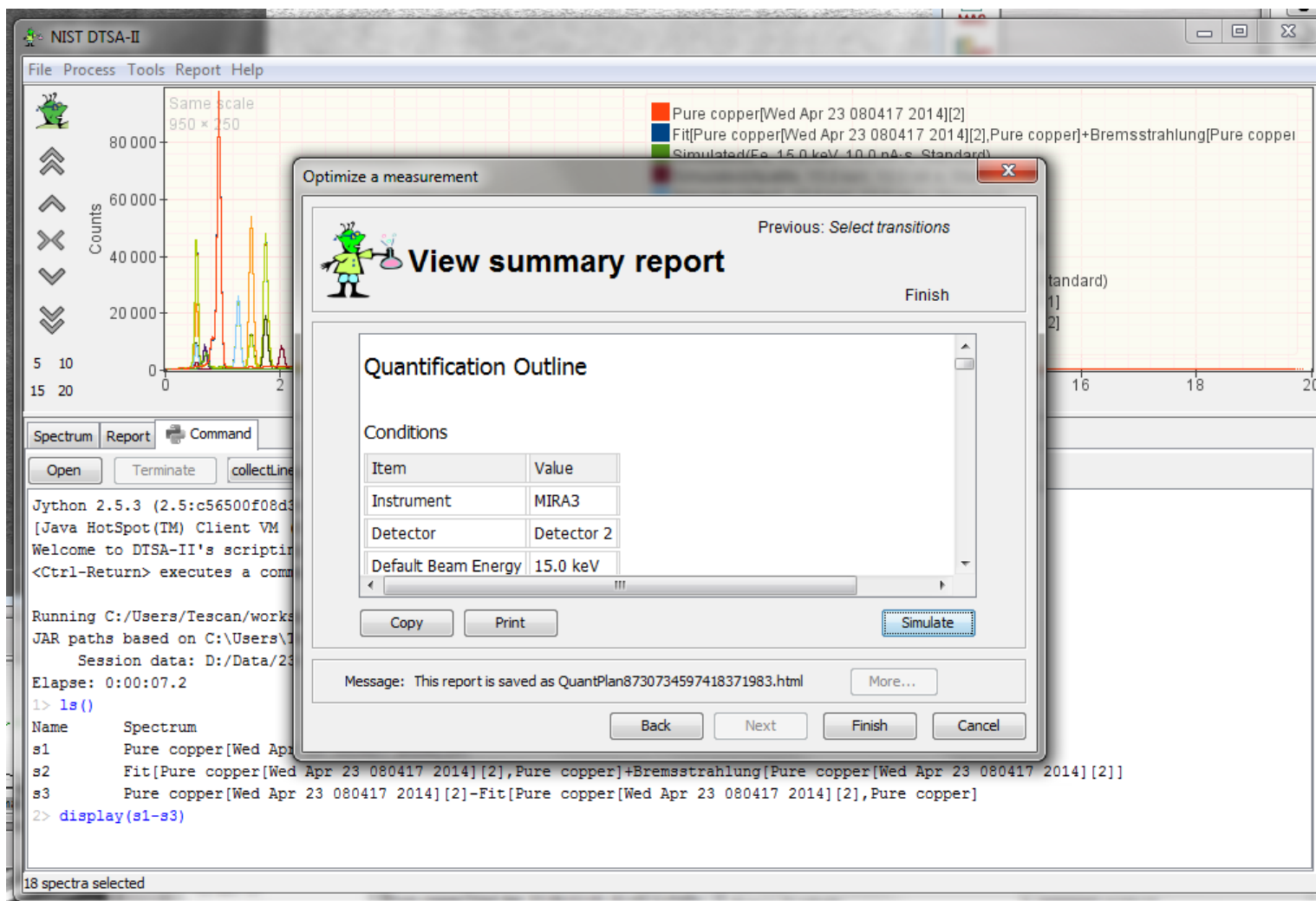






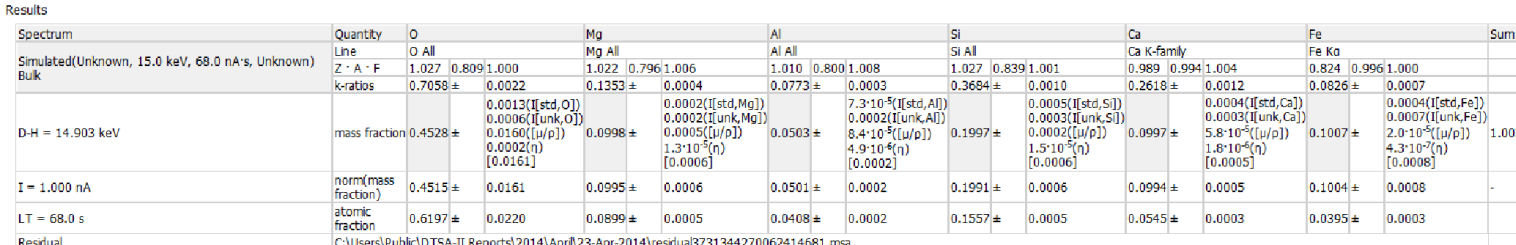






Measurement plan

Composition				Beam Energy (keV)	Form	Preparation	Dose (nA·s)	Type
Unknown				15.0	Bulk	Polished to a 50.0 nm finish	68.0	Unknown
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
O	0.4500	0.4500	0.6181					
Mg	0.1000	0.1000	0.0904					
Al	0.0500	0.0500	0.0407					
Si	0.2000	0.2000	0.1565					
Ca	0.1000	0.1000	0.0548					
Fe	0.1000	0.1000	0.0394					
CaF2				15.0	Bulk	Polished to a 50.0 nm finish	14.0	Reference
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
F	0.4867	0.4867	0.6667					
Ca	0.5133	0.5133	0.3333					
Al2O3				15.0	Bulk	Polished to a 50.0 nm finish	24.0	Reference & Standard
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
O	0.4707	0.4707	0.6000					
Al	0.5293	0.5293	0.4000					
MgO				15.0	Bulk	Polished to a 50.0 nm finish	10.0	Standard
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
O	0.3970	0.3970	0.5000					
Mg	0.6030	0.6030	0.5000					
SiO2				15.0	Bulk	Polished to a 50.0 nm finish	10.0	Standard
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
O	0.5326	0.5326	0.6667					
Si	0.4674	0.4674	0.3333					
Apatite				15.0	Bulk	Polished to a 50.0 nm finish	10.0	Standard
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
H	0.0019	0.0019	0.0435					
O	0.3900	0.3900	0.5652					
P	0.2323	0.2323	0.1739					
Ca	0.3758	0.3758	0.2174					
Fe				15.0	Bulk	Polished to a 50.0 nm finish	10.0	Standard
Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction					
Fe	1.0000	1.0000	1.0000					



Value
30 nm
250°
15.0 keV
614e610b-6ae5-6e57-3c48-b0eebf7b470
0 μm
Detector 2 - FWHM[Mn Kα] = 131.7 eV - F.
0570f03e-1acd-f1b1-80b6-c7aad8420556
30 mm ²
[0.280,0.770,0.574]
[-10.086,-27.711,-3.649]
0.45 mm
Silicon Drift Detector
Moxtek AP 3.3 (manufacturer's table)
35°
0.00000000 eV/ch ²
2.7 eV
10 eV/channel

Name	O	Mg	Al	Si	Ca	Fe
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)	45.28	9.98	5.03	19.97	9.97	10.07
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[1]	45.41	10.01	5.02	19.97	10.08	10.01
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[2]	45.00	9.94	4.99	19.93	10.03	9.90
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[3]	45.16	9.95	4.98	19.93	9.97	9.95
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[4]	45.06	9.96	5.02	19.92	9.96	9.92
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[5]	45.38	9.92	5.02	19.96	10.00	9.95
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[6]	45.10	9.96	5.00	19.92	10.00	9.82
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[7]	45.23	10.01	5.01	19.97	9.99	9.98
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[8]	45.27	9.98	5.02	19.95	10.02	10.06
Simulated(Unknown, 15.0 keV, 68.0 nA·s, Unknown)[9]	45.51	10.00	5.05	20.02	10.09	9.91
Average	45.24	9.97	5.01	19.95	10.01	9.96
Std. Dev.	0.1535	0.0274	0.0179	0.0297	0.0427	0.0702
Fractional	0.34%	0.27%	0.36%	0.15%	0.43%	0.71%